RAJESH YADAV GORLA

Mechanical Design Engineer | Siemens NX | CATIA V5 | GD&T | Wind-Tunnel Model & Motorsport Design
No visa sponsorship required | Full UK driving licence | Relocating to Milton Keynes
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PROFESSIONAL SUMMARY

Mechanical design engineer (MSc Automotive Engineering, Merit; BTech Aerospace Engineering, First Class) with 17 months of production design experience on Honda's EV programme (Honda 0 α), taking 10+ Body-in-White components from CATIA V5 design to engineering release and producing 100+ GD&T drawings under ENOVIA change control. Hands-on wind-tunnel and motorsport experience: led the CAD and FDM 3D printing of a 20% scale wind-tunnel model, designed a laser-cut aluminium steering rack mount fitted to the Coventry Formula Student car, and developed a CFD-verified aero package for a 750 Formula car. Trained in Siemens NX sheet-metal design and GD&T. Motivated to design precise, manufacturable wind-tunnel model hardware alongside aerodynamicists and the Model Shop. No visa sponsorship required.

CORE SKILLS

CAD: Siemens NX; CATIA V5 (Part Design, Assembly, Generative Shape Design, Drafting, FT&A); Autodesk Alias; 3D solid and surface modelling; assemblies; parametric modelling

Drawings & Tolerancing: Engineering drawings; GD&T; Functional Tolerancing & Annotation; tolerance and interface review; Engineering Change Requests; revision control; ENOVIA PLM

Analysis & Validation: FEA (ANSYS); CFD (Simcenter STAR-CCM+); mesh independence; CFD-to-wind-tunnel correlation; engineering calculations; Python (SciPy) optimisation; MATLAB/Simulink

Manufacturing & Materials: Design for manufacture and assembly; sheet metal (JIS-spec and mild steel); spot welding, hemming and adhesive bonding; laser-cut aluminium; FDM 3D printing; plastic part design

Design Process: Concept to release; QFD; Pugh matrix; FMEA; DFX; cross-functional working with design, manufacturing and quality teams; technical reporting

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer | L&T Technology Services – Client: Honda EV Programme | India
Mar 2024 – Aug 2025 | Full time

- Designed 10+ Body-in-White upper-body components (roof panels, roof bows, A/B/C pillars) in CATIA V5 for a vehicle since revealed as the Honda 0 α (Japan Mobility Show 2025), taking each from detailed design to engineering release.
- Produced and maintained 100+ GD&T engineering drawings using Functional Tolerancing & Annotation, building in manufacturing, assembly and quality requirements.
- Designed JIS-specification and mild steel sheet-metal parts for spot welding, hemming and adhesive bonding, reviewing interfaces and clearances with surrounding parts.
- Raised and managed Engineering Change Requests in ENOVIA, keeping models and drawings under revision control and verifying designs before release.
- Resolved component, interface and drawing issues within a 15-member cross-functional team to programme milestones; owned first-issue ("birth") drawing release and guided colleagues on drawing practice.

CAD Engineer (Paid Internship, Full time) | Invictus Defence Systems | Bengaluru, India
Apr 2022 – Feb 2024

- Modelled and revised defence UAV airframe geometry in CATIA V5, implementing design changes identified through ANSYS airflow analysis.

ENGINEERING PROJECTS – WIND TUNNEL, MOTORSPORT & DESIGN

DrivAer Wind-Tunnel Model and CFD Study | Coventry University | Group lead, team of 5
MSc Aerodynamics module | Apr 2026

- Led the team that CAD-modelled and FDM 3D-printed a 20% scale DrivAer model for wind-tunnel testing, instrumented with 62 surface-pressure taps.
- Planned the test programme and split the analysis: fastback, notchback and estate rear ends with smooth and detailed underbodies, tested for forces, Reynolds-number sweeps, ±12° yaw and wool-tuft flow visualisation.
- Built a 3.63-million-cell polyhedral STAR-CCM+ model (mesh independence within 2% Cd) that matched tunnel Cd within 5.1% and CL within 1.1%.
- Found the smooth underbody cut drag by 5.3% but raised lift by 27%; recommended keeping the detailed underbody and adding a front splitter to bring lift towards neutral.

Steering Rack Mount | Coventry University Formula Student Team
Sep 2025 – Sep 2026

- Designed a new steering rack mount in CATIA V5 around the car's existing packaging and interfaces (PR25-T reference geometry), revised to meet Formula Student technical regulations.
- Detailed the part for laser cutting in aluminium; the mount was manufactured and fitted to the team car.

750 Formula Aero Package – MSc Dissertation | Coventry University (real 750 Motor Club race car; CAD supplied by the team)
Apr 2026 – Aug 2026

- Found and reviewed 452 surface-quality problem regions in the supplied full-vehicle CAD before meshing, then ran a 13-level mesh study (1.1–7.6 million cells) to select a 6.4-million-cell working mesh in Simcenter STAR-CCM+.
- Designed a single-element NACA 0012 rear wing, diffuser and wheel-arch inlets in CATIA V5 within 750 Formula size and height limits, verifying one change at a time in CFD.
- Cut drag coefficient by 9.9% (0.537 to 0.484) and drag force by about 8% (771 N to 708 N at 50 m/s), despite a 2% larger frontal area.
- Reported the trade-off openly (downforce coefficient reduced by about 47%) and recommended the next step: minimum drag at a fixed downforce and axle balance.

Lightweight UAV Propeller Guard – Design and Optimisation | Coventry University
MSc Automotive Engineering, 2025 – 2026

- Took a 3D-printed propeller guard from requirements to final design using QFD, morphological analysis, Pugh matrix, FMEA and DFX.
- Built a parametric CATIA V5 model, assessed impact (50 N), flight (10 N) and crush (100 N) load cases in FEA, and wrote a Python (SciPy) optimisation of rib count, rib thickness and diameter that cut mass by 23% within stress and deflection limits.

EDUCATION

MSc Automotive Engineering (Merit) | Coventry University, UK | 2025 – 2026
Modules: Generative Shape Design; Structural Integrity and Dynamic Behaviour; Aerodynamics and Vehicle Dynamics; Future Automotive Technologies.

BTech Aerospace Engineering (First Class) | SASTRA Deemed University, India | 2018 – 2022
Modules: Aerodynamics; Finite Element Analysis; Computational Fluid Dynamics.

TRAINING & CERTIFICATES

- **Automotive Sheet Metal Design using Siemens NX CAD / CATIA V5** – Skill-Lync, Jul 2023 (bonnet, side sill and roof panel projects)
- **Geometric Dimensioning and Tolerancing** – Skill-Lync, Jul 2023
- **Automotive Plastic Design using CATIA V5** – Skill-Lync, May 2023

ADDITIONAL INFORMATION

- **Right to work:** Student visa valid to January 2027 with full-time work permitted; applying for the UK Graduate Route in December 2026 (2 years, no sponsorship required). Open to Skilled Worker sponsorship if offered.
- **Mobility:** Full UK driving licence; willing to relocate to Milton Keynes and support travel when required.
- **Languages:** English (fluent); German and Japanese (beginner).